

Theory of Automata

Assignment No. 3

Submit the Hard copy to your Instructor's office (or In class) before 5:00 pm on 20th November.

Due Date: 20 - November -2025

Question # 1: Give a context-free grammar (CFG) for each of the following languages over the alphabet $\Sigma = \{a, b\}$.

- a. $L = \{ a^n b^m \mid n > 0, 2n \leq m \leq 3n \}$
- b. $L = \{ (ab)^n (ab^m)^n \mid n \geq 0, m > 0 \}$
- c. String doesn't contain abba
- d. CFG of w Length strings, where w is the last number of your roll number.

Question #2 Remove λ productions from these grammars.

- a. $S \rightarrow aC \mid bC$
 $C \rightarrow a \mid b \mid \lambda$
- b. $S \rightarrow aC \mid bS \mid a \mid b$
 $C \rightarrow aC \mid a \mid \lambda$
- c. $S \rightarrow YaY \mid bY$
 $Y \rightarrow YaY \mid YbY \mid \lambda$

Question #3 Remove unit productions.

- a. $S \rightarrow CD$
 $C \rightarrow D$
 $D \rightarrow aD \mid Db \mid \lambda$
- b. $S \rightarrow AA$
 $A \rightarrow B \mid BB$
 $B \rightarrow aB \mid Bb \mid \lambda$

Question #4 Convert the following CFG into CNF. You must show all the steps.

- a. $S \rightarrow aSa \mid SSa \mid a$
- b. $S \rightarrow AaB \mid c$
 $B \rightarrow b$
 $A \rightarrow a$
 $C \rightarrow aC$

c. $S \rightarrow aS \mid Z$

$Z \rightarrow a \mid C$

$C \rightarrow b$

d. $S \rightarrow SaS \mid SaSbS \mid SbSaS \mid \lambda$

Question#5 Convert the following CNF from question 4 to GNF. You must show all the steps.